Find My Stuff : A Lost And Found Management System

A PROJECT REPORT

Submitted by

N.Shirisha **23331A0744**

In the fulfilment for the

Community Service Project of

BACHELOR OF TECHNOLOGY

Under the Esteemed Guidance of

Mrs. S B V VaraLakshmi

At

DEPARTMENT OF INFORMATION ENGINEERING &

COMPUTATIONAL TECHNOLOGY

MAHARAJ VIJAYARAM GAJAPATHI RAJ COLLEGE OF ENGINEERING (AUTONOMOUS)

VIZIANAGARAM-535005, AP (INDIA)

DECLARATION

I hereby declare that the project entitled **"MULTILEVEL AUTHENTICATION SYSTEM"** submitted for the fulfilment of CSP Project in B.Tech Degree is my original work and the project has not formed on the basis for the submission in any degree or any other similar titles.

N.Shirisha(23331AO744)

Place :

Date :

CERTIFICATE

This is to certify that the project entitled **"MULTILEVEL AUTHENTICATION SYSTEM"** is the bonafide work carried out by N.Shirisha(23331A0744) of B.tech V Sem IECT, MVGR College of Engineering (Autonomous), Vizianagaram, during the year 2024-2025, in fulfilment of the Mini project in Bachelor of Technology and that project has not formed the basis for the submission previously of any degree or any other similar title.

Signature of Project Guide

Mrs. S B V VaraLakshmi (M.Tech)
Assistant Professor
Dept of IECT
MVGR College of Engineering(A)

Signature of Head of the Department

Dr. Anjana Devi .B (M.Tech, PhD)
Associate Professor & HOD
Dept. of IECT
MVGR College of Engineering(A)

Place :

Date :

TABLE OF CONTENTS

Chapter	Title	Page No.
1	ABSTRACT	1
2	INTRODUCTION	2
3	PROBLEM STATEMENT	3
4	SYSTEM REQUIREMENTS	4
	4.1 HARDWARE REQUIREMENTS	
	4.2 SOFTWARE REQUIREMENTS	
5	TECHNOLOGIES USED	5-9
6	EXISTING SYSTEM	10
7	PROPOSED SYSTEM	11
8	USE-CASES	12-13
9	PROCESS-FLOW	14
10	SAMPLE CODE	20
11	OUTPUT SCREENSHOTS	24
12	ADVANTAGES	25
13	LIMITATIONS	26
14	CONCLUSION	27
15	FUTURESCOPE	28
16	REFERENCE	29

1. ABSTRACT

The Find My Stuff project is a web-based application designed to simplify the process of reporting and recovering lost or found items. In many real-life situations, people lose valuable belongings in public places, and it becomes difficult to trace them back. This system provides a digital platform where users can post details of lost or found items along with their location, description, and images. The application is built using the Flask framework for the backend and HTML, CSS, and JavaScript for the frontend. Data is securely stored in an SQLite database, ensuring efficient retrieval and management of item records. The system enhances transparency, saves time, and promotes community cooperation by connecting people who have lost items with those who have found them. Overall, Find My Stuff serves as a smart and user-friendly solution to bridge the gap between lost and found cases through technology.

2. INTRODUCTION

In today's fast-paced world, losing personal belongings has become a common problem. People often misplace their items in public places such as colleges, offices, buses, or parks and face great difficulty in retrieving them. Similarly, individuals who find lost items have no proper platform to report or return them to the rightful owners. The **Find My Stuff** project is developed to solve this real-world issue through a simple and efficient web-based system.

This application allows users to **report lost items** or **upload details of found items** along with essential information such as item name, description, location, date, and image. The system stores this data in a centralized **SQLite database** and provides a **user-friendly dashboard** to view and manage all entries.

The project uses the **Flask framework** as the backend server, which handles all application logic, form processing, and database operations. The **frontend** is built using **HTML, CSS, and JavaScript**, providing a smooth and interactive interface for users. Flask templates (Jinja2) dynamically render the web pages, connecting the backend data with the visual interface.

The main objective of this project is to **create a bridge between people who lose items and those who find them**, reducing the communication gap and making the recovery process faster and more transparent. With features like data validation, image uploads, and organized storage, **Find My Stuff** ensures both accuracy and convenience.

This system not only saves time but also encourages honesty and community participation by helping users return lost belongings easily. Hence, **Find My Stuff** serves as a smart, secure, and reliable platform to digitally manage lost-and-found activities in a modern, technology-driven environment.

3. PROBLEM STATEMENT

In everyday life, people often lose or misplace their personal belongings such as mobile phones, wallets, ID cards, books, or accessories in public places like colleges, offices, and transportation areas. Likewise, many people find lost items but have no proper medium to return them to the rightful owners. The existing manual process of finding and claiming lost items is inefficient, time-consuming, and lacks transparency. There is **no centralized platform** that allows users to easily report, search, or manage lost and found items. As a result, many valuable belongings remain unclaimed or are permanently lost. To address this issue, there is a need for a **digital system** that can record, manage, and match lost and found data efficiently.

The **Find My Stuff** project aims to solve this problem by providing an **online web-based application** where users can upload, search, and retrieve information about lost or found items with location, description, and images. This system bridges the gap between people who lose items and those who find them, ensuring a faster and more reliable recovery process.

4. SYSTEM REQUIREMENTS

4.1 HARDWARE SPECIFICATION:

- **Operating System:** Windows 10 or above
- **Web Framework:** Flask
- **Language:** Python 3.x
- **Database:** SQLite
- **Browser:** Chrome, Firefox, or Edge

4.2 SOFTWARE SPECIFICATION:

- **Hard Disk:** 100 MB free space
- **RAM:** 4 GB or higher
- **Processor:** Intel i3 or higher

5. TECHNOLOGIES USED

1. PYTHON:

Python is a powerful, high-level, and easy-to-understand programming language used as the core backend language for this project. It supports object-oriented and modular programming, making code easier to maintain and extend.

In the Find My Stuff project, Python is used to:

- Connect the Flask framework to the SQLite database.
- Handle routes and form submissions.
- Process user input, such as lost and found item data.
- Perform server-side validation and logical operations.

Its rich library support and simplicity make Python an ideal choice for rapid web application development.

2. Flask Framework:

Flask is a micro web framework built in Python. It is lightweight but highly extensible, which makes it perfect for small to medium-sized web applications. Flask provides simplicity and flexibility while maintaining high performance .

In this project, Flask acts as the backend server, managing all the communication between the frontend and the database.

Main roles of Flask in this project:

- Handles URL routing (@app.route) to different pages.
- Connects to the SQLite database to store and fetch item details.
- Renders HTML templates dynamically using Jinja2.
- Manages sessions, user requests, and responses.
- Displays flash messages for success or error notifications.

3.HTML (HyperText Markup Language)

HTML is the **standard markup language** used to create the structure of all web pages. It defines how the content is displayed on the browser.
In the **Find My Stuff** project:

- HTML forms are used for entering lost or found item details.
- Pages like **Home**, **Dashboard**, **Lost Item Form**, and **Found Item Form** are built using HTML elements.
- It acts as the skeleton of the website, which is later styled using CSS and made interactive using JavaScript.

HTML ensures that all visual elements are properly organized for the user interface.

4.CSS (Cascading Style Sheets)

CSS is used to **style and beautify** the web pages created with HTML. It allows customization of colors, fonts, layouts, and animations.

In this project:

- CSS provides a **modern and appealing design** for the dashboard and forms.
- It uses **gradients, transitions, and hover effects** to improve the look and feel.
- Layouts are made responsive to fit on laptops and mobile devices.

CSS makes the project visually attractive and enhances user experience.

5. JavaScript

JavaScript is a **client-side scripting language** that adds interactivity and dynamic behavior to web pages.

In the **Find My Stuff** system:

- It is used to show or hide sections dynamically (like switching between “Browse” and “Claim” views).
- Performs **client-side validations**, ensuring correct user input before form submission.
- Enhances user experience by reducing unnecessary page reloads.

JavaScript brings life to the website by making it more engaging and interactive.

6. SQLite Database

SQLite is a **lightweight relational database** used to store all the project's data locally. It doesn't require a separate server and is built into Python.

In this project:

- It stores all lost and found item details (name, description, location, date, image).
- It allows CRUD operations — **Create, Read, Update, Delete**.
- Data retrieval is fast and efficient, making it suitable for a web-based platform.

SQLite ensures secure, stable, and reliable data management.

7. Bootstrap (Frontend Framework)

Bootstrap is a **CSS framework** used to create responsive and mobile-friendly web designs. It comes with pre-built templates, buttons, forms, and layouts.

In this project:

- Bootstrap simplifies the design process and ensures consistency across pages.
- It is used to create attractive navigation bars, buttons, and forms with minimal code.
- It automatically adjusts layouts for mobile, tablet, and desktop screens.

Bootstrap saves time and improves the overall UI/UX quality.

8. Jinja2 Template Engine

Jinja2 is a **templating engine** used with Flask to render dynamic HTML pages.

In this project:

- It allows Python variables (like item name, location, and images) to be displayed in HTML using {{ }}.
- Helps create dynamic dashboards and item lists directly from the database.
- Reduces code duplication by using reusable templates (like header and footer sections).

Jinja2 connects the backend data with the frontend interface seamlessly.

6.EXISTING SYSTEM

- In the current real-world scenario, the process of reporting and recovering lost items is mostly **manual and unorganized**. People usually rely on physically searching in public places, asking others, or posting on social media groups to find lost belongings.
- Similarly, when someone finds a lost item, there is no proper centralized platform to report it, making it difficult to return the item to the rightful owner.
- The existing system is **inefficient and time-consuming**. There is no official record of lost or found items, which often leads to items being permanently lost or remaining unclaimed.
- Communication between the person who lost an item and the person who found it is unreliable and slow. Moreover, there is no verification or validation, which can result in misinformation or disputes regarding ownership.
- Due to these challenges, there is a clear need for a **digital, systematic approach** that can efficiently track lost and found items, allow users to upload item details, and help match items with their rightful owners.

6. PROPOSED SYSTEM

- In the current real-world scenario, the process of reporting and recovering lost items is mostly **manual and unorganized**. People usually rely on physically searching in public places, asking others, or posting on social media groups to find lost belongings.
- Similarly, when someone finds a lost item, there is no proper centralized platform to report it, making it difficult to return the item to the rightful owner.
- The existing system is **inefficient and time-consuming**. There is no official record of lost or found items, which often leads to items being permanently lost or remaining unclaimed.
- Communication between the person who lost an item and the person who found it is unreliable and slow. Moreover, there is no verification or validation, which can result in misinformation or disputes regarding ownership.
- Due to these challenges, there is a clear need for a **digital, systematic approach** that can efficiently track lost and found items, allow users to upload item details, and help match items with their rightful owners.

7. USE CASES

1. User Registration and Login

- **Description:** Users can create a new account by providing their name, email, and password. Registered users can log in to access the system features.
- **Purpose:** Ensures only verified users can report or claim items, maintaining security and accountability.

2. Report Lost Item

- **Description:** Users can submit details about an item they have lost, including the item name, description, location, date, and image.
- **Purpose:** Helps create a record of lost items in the system so that others can view and identify them if found.

3. Report Found Item

- **Description:** Users who find an item can submit its details, such as description, location, date, and image.
- **Purpose:** Ensures found items are logged into the system and can be matched with lost item reports.

4. Browse Lost and Found Items

- **Description:** Users can view a list of lost or found items through a dashboard. The system displays all relevant information along with images.
- **Purpose:** Helps users quickly search for their lost items or identify items they have found.

5. Claim Item

- **Description:** Users can claim a found item by contacting the person who reported it. The system may provide contact details or notifications.
- **Purpose:** Facilitates the smooth return of items to their rightful owners.

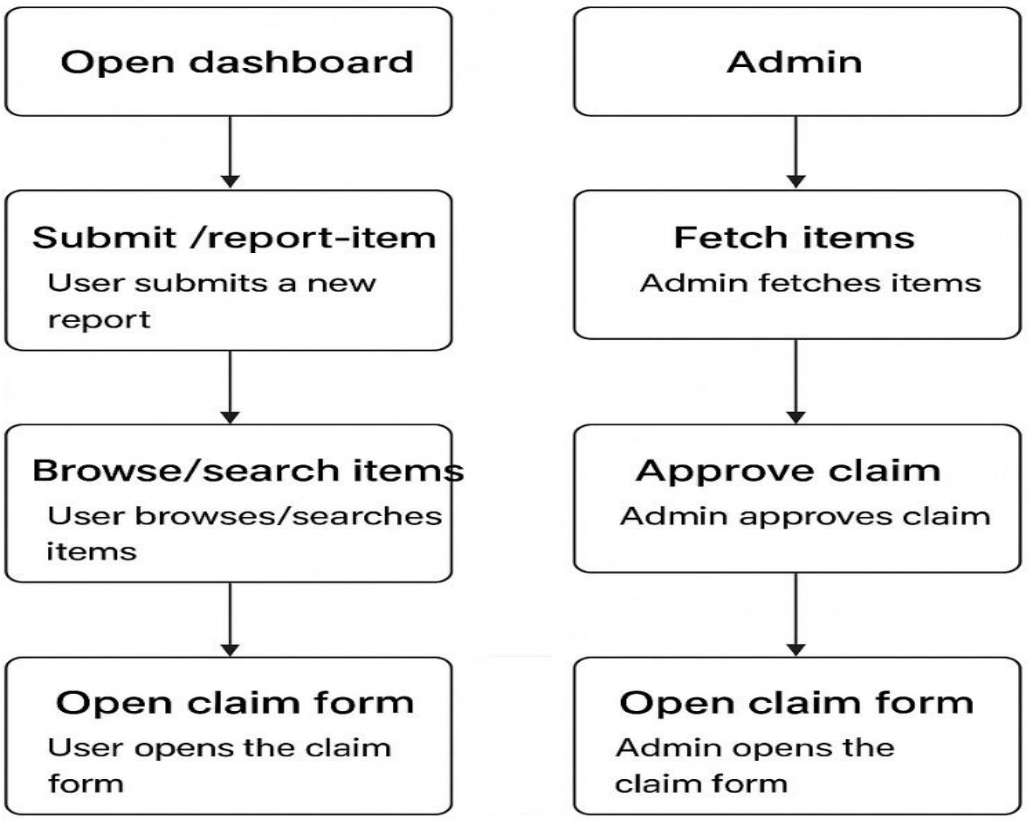
6. Admin Dashboard (Optional)

- **Description:** An admin can manage user accounts, monitor lost/found items, and remove fake or inappropriate entries.
- **Purpose:** Maintains the reliability, integrity, and accuracy of the system.

7. Notifications (Optional)

- **Description:** Users may receive notifications when their lost item is found or when a claimed item is updated.
- **Purpose:** Keeps users informed and improves the efficiency of item recovery.

8. PROCESS FLOW



9. SAMPLE CODE

MAIN CODE :

```
// ----------------- SHOW / HIDE SECTIONS -----------------
function showSection(sectionId) {
   const sections = ["dashboard-container", "browse", "claimSection", "myClaims"];

   // Hide all sections
   sections.forEach(id => {
      const el = document.getElementById(id);
      if (el) el.style.display = "none";
   });

   // Show the requested section
   const target = document.getElementById(sectionId);
   if (target) target.style.display = "block";
}

// ----------------- CLOSE MODALS FUNCTIONS -----------------
function closeReplyModal() {
   document.getElementById('replyModal').style.display = 'none';
}

function closeModal(modalId) {
   document.getElementById(modalId).style.display = 'none';
}

// ----------------- DOM CONTENT LOADED -----------------
document.addEventListener("DOMContentLoaded", function () {

   // ----------------- GET MAIN ELEMENTS -----------------
   const dashboard = document.getElementById("dashboard-container");
   const browseSection = document.getElementById("browse");
   const claimSection = document.getElementById("claimSection");
   const myClaimsSection = document.getElementById("myClaims");
   const reportModal = document.getElementById("reportModal");

   // Hide Browse, Claim, My Claims initially
   if (browseSection) browseSection.style.display = "none";
   if (claimSection) claimSection.style.display = "none";
   if (myClaimsSection) myClaimsSection.style.display = "none";

   // ----------------- REPORT MODAL -----------------
   const lostTab = document.getElementById("lostTab");
   const foundTab = document.getElementById("foundTab");
   const formTitle = document.getElementById("formTitle");

   window.openModal = function(type) {
      if (reportModal) reportModal.classList.add("show");
      switchTab(type === "lost" ? "lost" : "found");
```

```
    const reportForm = document.getElementById("reportForm");
    if (reportForm) reportForm.reset();
    if (dashboard) dashboard.style.filter = 'blur(5px)';
};

window.closeModal = function() {
    if (reportModal) reportModal.classList.remove("show");
    if (dashboard) dashboard.style.filter = 'none';
};

window.switchTab = function(tabName) {
    if (!lostTab || !foundTab || !formTitle) return;
    const itemKindInput = document.getElementById("itemKind");
    const reportForm = document.getElementById("reportForm");
    if (reportForm) reportForm.reset();

    if (tabName === "lost") {
        lostTab.classList.add("active");
        foundTab.classList.remove("active");
        formTitle.textContent = "Report Lost Item";
        if (itemKindInput) itemKindInput.value = "Lost";
    } else {
        lostTab.classList.remove("active");
        foundTab.classList.add("active");
        formTitle.textContent = "Report Found Item";
        if (itemKindInput) itemKindInput.value = "Found";
    }
};

// ----------------- BROWSE SECTION -----------------
window.openBrowsePage = function() {
    showSection('browse');
};

window.backToDashboard = function() {
    showSection('dashboard-container');
};

window.filterItems = function() {
    const searchValue = document.getElementById("searchBox").value.toLowerCase();
    const categoryValue = document.getElementById("categoryFilter").value.toLowerCase();
    const items = document.querySelectorAll("#itemsGrid .step");

    items.forEach(item => {
        const title = item.dataset.title.toLowerCase();
        const description = item.dataset.description.toLowerCase();
        const location = item.dataset.location.toLowerCase();
        const itemCategory = item.dataset.category.toLowerCase();

        if ((title.includes(searchValue) || description.includes(searchValue) ||
location.includes(searchValue) || searchValue === "") &&
```

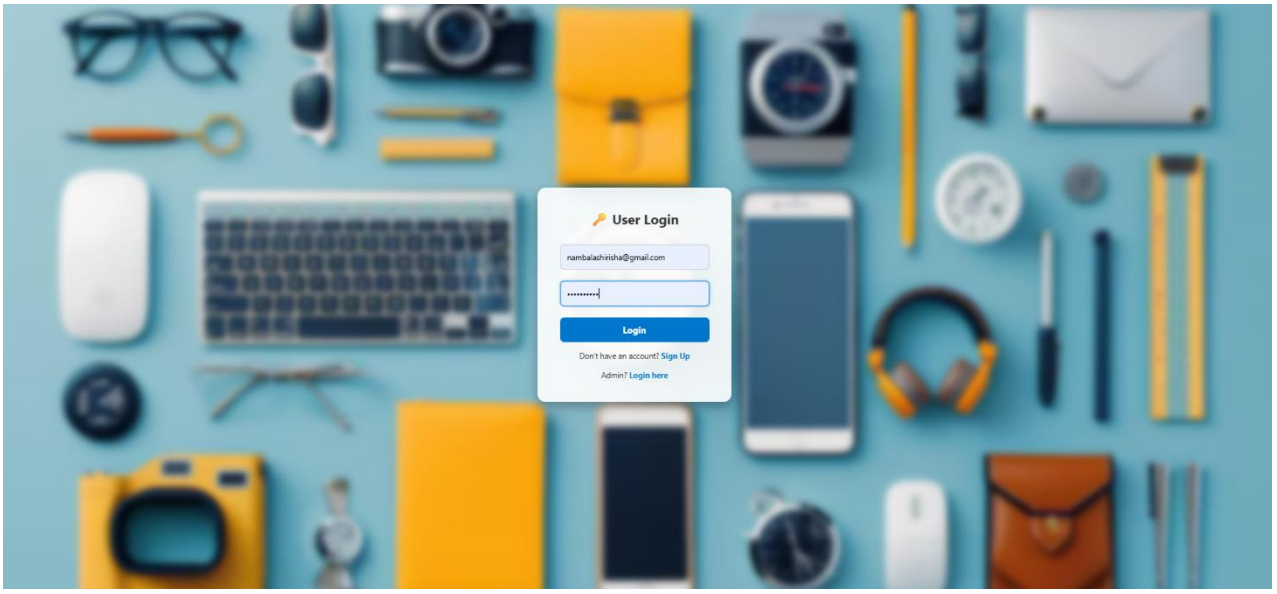
```
            (itemCategory === categoryValue || categoryValue === "")) {
            item.style.display = "block";
        } else {
            item.style.display = "none";
        }
    });
};

// ----------------- CLAIM SECTION -----------------
const claimForm = document.getElementById("claimForm");
if (claimForm) {
    claimForm.addEventListener("submit", function(e) {
        e.preventDefault();
        const message = document.getElementById("message").value.trim();
        if (!message) {
            alert("Please fill all fields");
            return;
        }
        claimForm.submit();
    });
}

document.querySelectorAll('.claim-btn').forEach(btn => {
    btn.addEventListener('click', () => {
        document.getElementById('claimItemId').value = btn.dataset.itemId;
        document.getElementById('claimItemTitle').textContent = btn.dataset.itemTitle;
        document.getElementById('claimItemDescription').textContent =
btn.dataset.itemDescription;
        document.getElementById('claimItemLocation').textContent = btn.dataset.itemLocation;
        showSection('claimSection');
    });
});

const backToBrowseBtn = document.getElementById('backToBrowse');
if (backToBrowseBtn) {
    backToBrowseBtn.addEventListener('click', () => {
        showSection('browse');
    });
}

window.closeClaimModal = function() {
    showSection('dashboard-container');
};

// ----------------- REPORT FORM VALIDATION -----------------
const reportForm = document.getElementById("reportForm");
if (reportForm) {
    reportForm.addEventListener("submit", function(e) {
        const title = reportForm.title.value.trim();
        const category = reportForm.category.value.trim();
        const date = reportForm.date.value.trim();
```

```
        const description = reportForm.description.value.trim();
        const location = reportForm.location.value.trim();
        const contact_email = reportForm.contact_email.value.trim();
        const contact_phone = reportForm.contact_phone.value.trim();

        if (!title || !category || !date || !description || !location || !contact_email || !contact_phone) {
            e.preventDefault();
            alert("Please fill all required fields including contact info");
        }
    });
}

// ----------------- REPLY MODAL -----------------
document.querySelectorAll('.reply-btn').forEach(btn => {
    btn.addEventListener('click', () => {
        const claimId = btn.dataset.claimId;
        document.getElementById('replyClaimId').value = claimId;
        document.getElementById('replyModal').style.display = 'flex';
    });
});

document.getElementById('closeReplyModal').addEventListener('click', closeReplyModal);
document.getElementById('cancelReply').addEventListener('click', closeReplyModal);

// ----------------- REQUEST INFO MODAL -----------------
document.querySelectorAll('.info').forEach(btn => {
btn.addEventListener('click', () => {
    const claimId = btn.dataset.claimId;
    document.getElementById('requestInfoClaimId').value = claimId;
    document.getElementById('requestInfoModal').style.display = 'flex';

    // Update status immediately
    fetch(`/update-status/${claimId}`, {
method: 'POST',
headers: { 'Content-Type': 'application/json' },
body: JSON.stringify({ status: 'Need Info' })
})
.then(res => res.json())   // <-- add this line to parse JSON response
.then(data => {
    if (data.success) {     // <-- check if backend responded with success
        const statusSpan = btn.closest('tr').querySelector('.status');
        if (statusSpan) {
            statusSpan.textContent = 'Need Info';
            statusSpan.className = 'status need-info';
        }
    } else {
        alert('Status update failed');
    }
})
.catch(err => console.error(err));
```

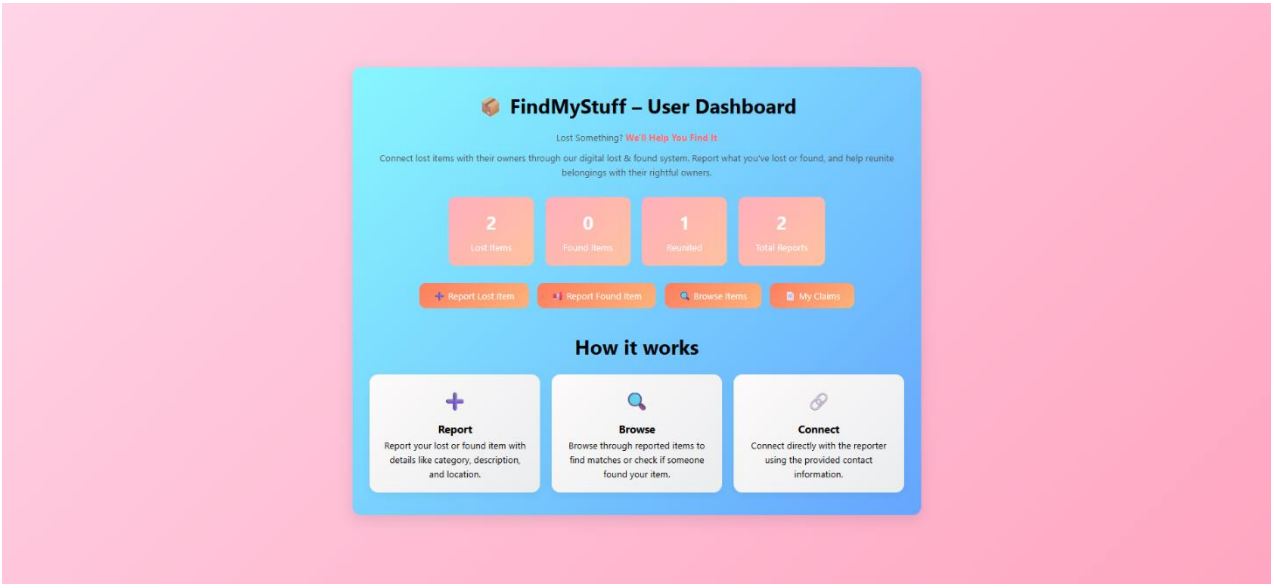
```
    });
});


    // ----------------- OUTSIDE CLICK HANDLER -----------------
    window.addEventListener('click', function(event) {
        const replyModal = document.getElementById('replyModal');
        const requestInfoModal = document.getElementById('requestInfoModal');
        const reportModal = document.getElementById('reportModal');
        const claimSection = document.getElementById('claimSection');

        if (event.target === replyModal) closeReplyModal();
        if (event.target === requestInfoModal) closeModal('requestInfoModal');
        if (event.target === reportModal) closeModal();
        if (event.target === claimSection) closeClaimModal();
    });
const backFromMyClaimsBtn = document.getElementById('backFromMyClaims');
if (backFromMyClaimsBtn) {
    backFromMyClaimsBtn.addEventListener('click', () => {
        showSection('dashboard-container');
    });
}
setTimeout(() => {
  const alerts = document.querySelectorAll('.alert');
  alerts.forEach(alert => alert.remove());
}, 4000);

});
```

10. OUTPUT SCREENSHOTS

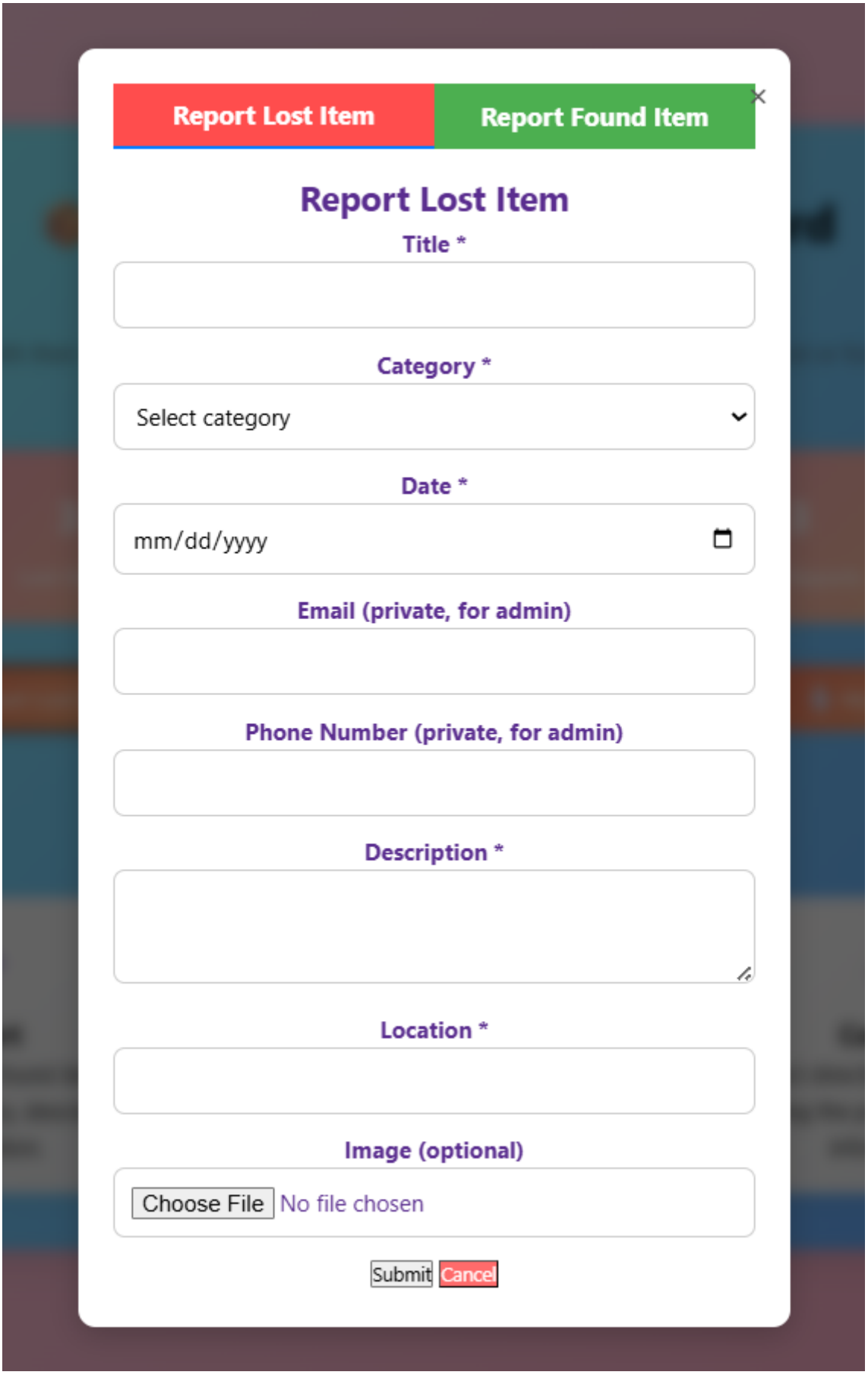
10.1 Home Page:



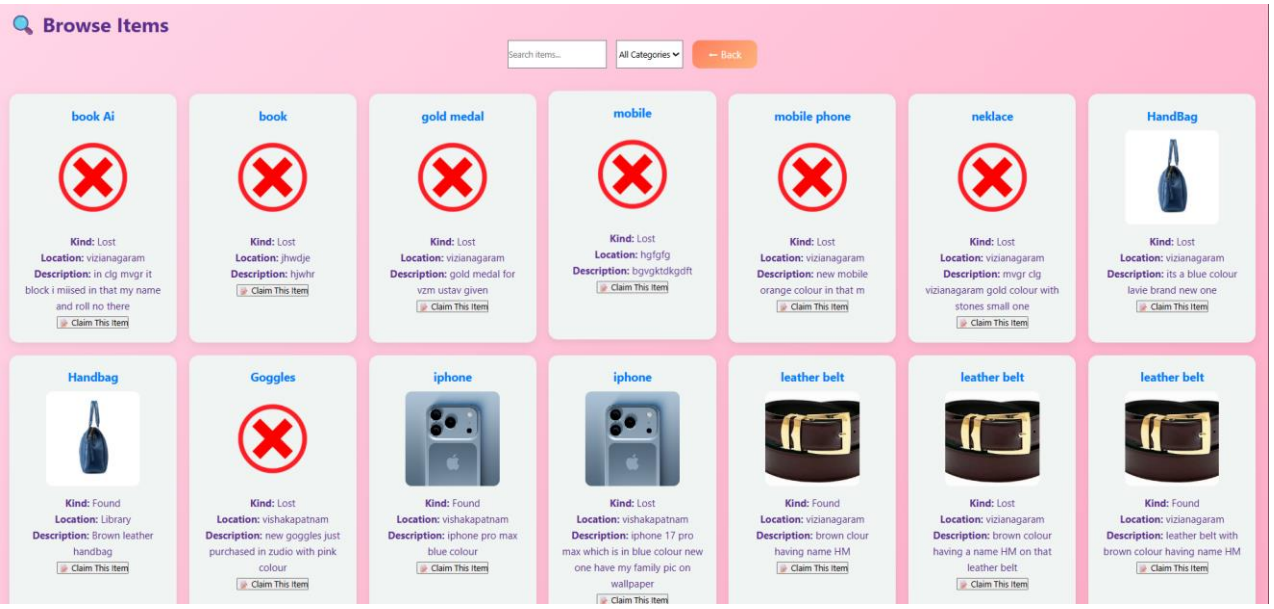
10.2 User Dashboard page:



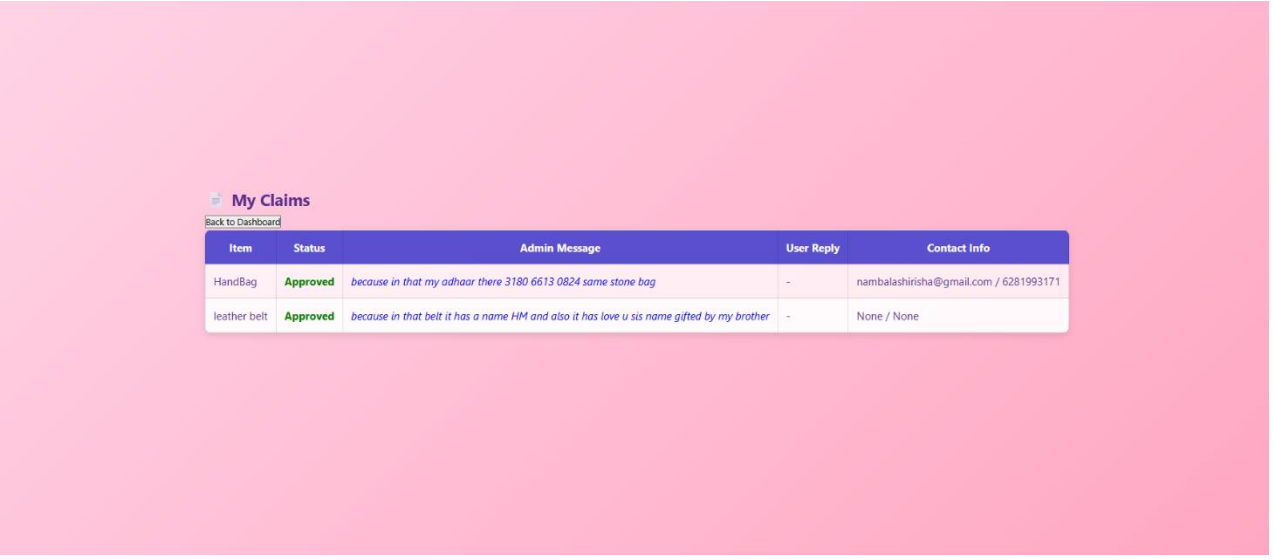
10.3 Lost and Found Page



10.4 Browse Page:

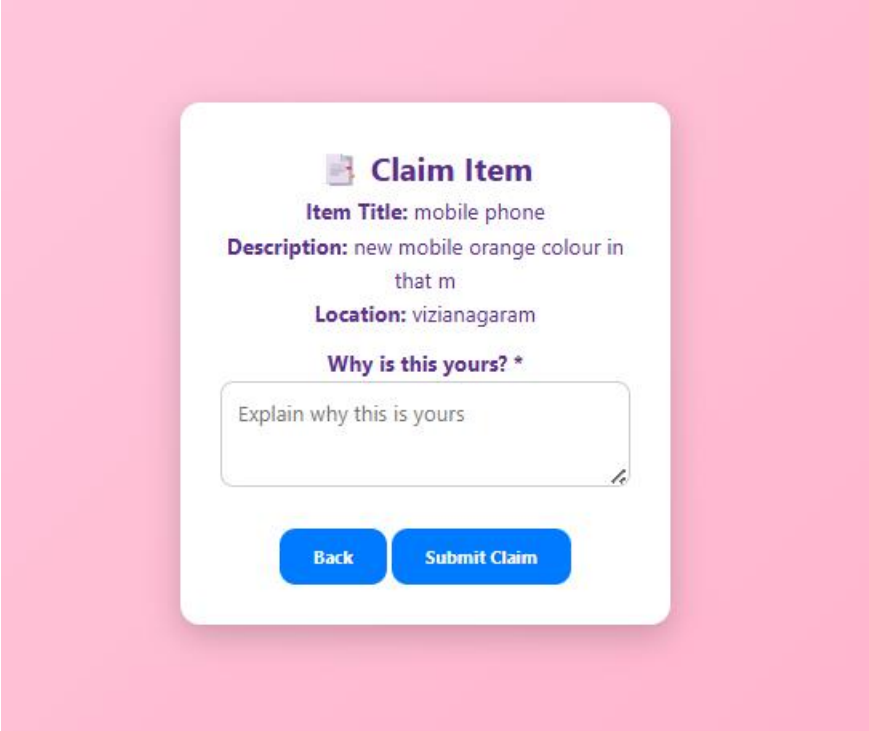


10.5 MyClaims:

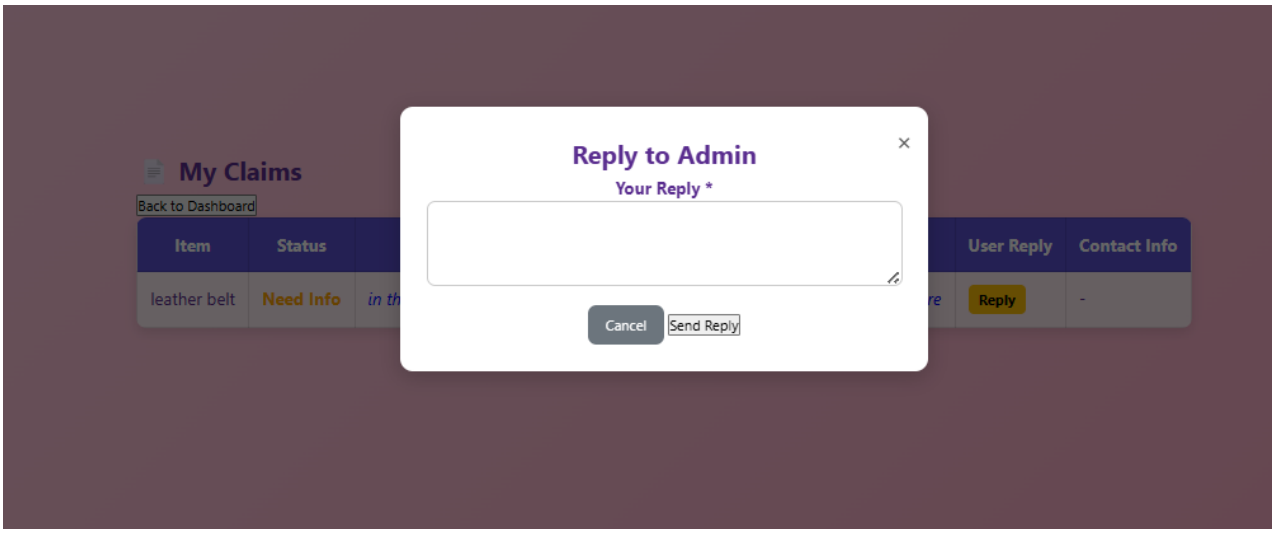


Item	Status	Admin Message	User Reply	Contact Info
HandBag	Approved	*because in that my adhaar there 3180 6613 0824 same stone bag*	-	nambalashirisha@gmail.com / 6281993171
leather belt	Approved	*because in that belt it has a name HM and also it has love u sis name gifted by my brother*	-	None / None

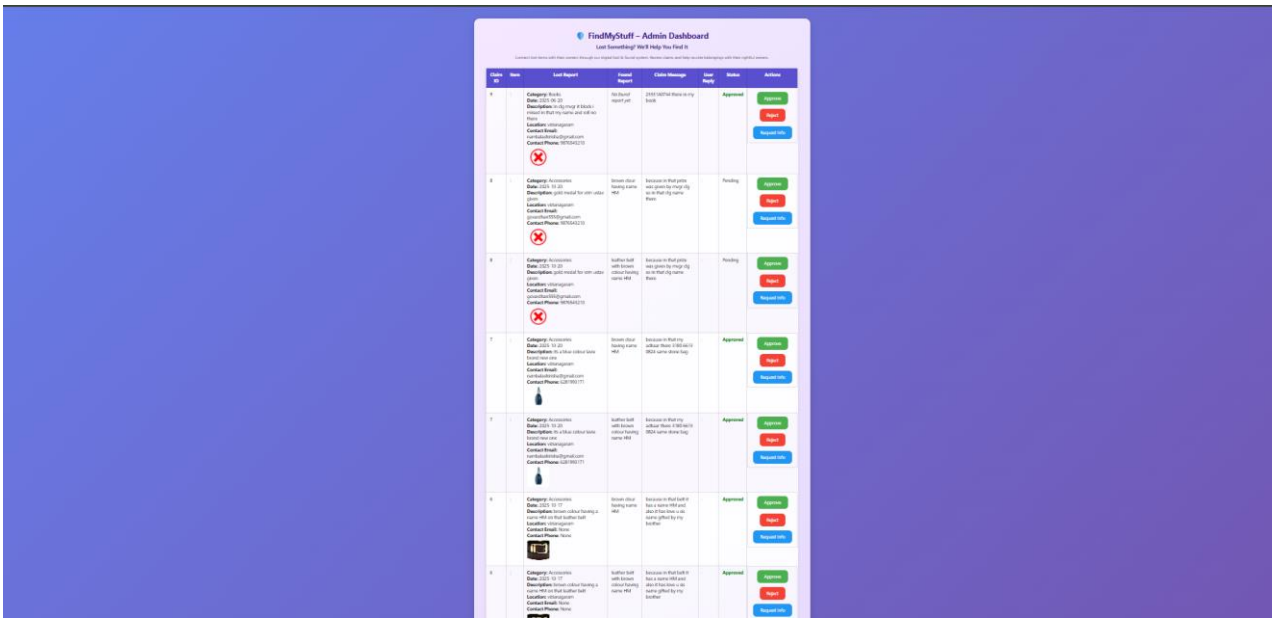
10.6 Claim Item:



10.7 User Reply:



10.8 Admin Dashboard Page:



11.ADVANTAGES

1. **Time-Saving:** The system allows users to report and search for lost items quickly without physically searching in multiple places.

2. **Centralized Platform:** All lost and found items are recorded in a single database, making it easy to access and manage information.

3. **User-Friendly Interface:** The dashboard and forms are designed to be simple and intuitive, so users can easily submit or browse item details.

4. **Enhanced Communication:** Users can connect with the person who found or lost an item, improving the chances of recovering lost belongings.

5. **Data Security:** The system uses user registration and validation to maintain the authenticity and integrity of item reports.

6. **Visual Representation:** Images of lost and found items help users quickly identify their belongings.

7. **Reduced Dependency on Manual Methods:** Eliminates the need for public notices, social media posts, or physical announcements, which are often inefficient.

8. **Accessibility:** Being a web-based application, it can be accessed from anywhere using a web browser.

12.LIMITATIONS

1. **Internet Dependency:** The system requires an internet connection to access the web application. Without it, users cannot report or search for items.

2. **Limited to Registered Users:** Only users with an account can access full features; new users need to register before using the system.

3. **Manual Verification Needed:** Although users can report items, verifying the authenticity of lost or found items may still require manual confirmation.

4. **Database Storage Limitations:** SQLite is suitable for small to medium projects but may face performance issues with very large datasets.

5. **No Real-Time Location Tracking:** The system relies on user-inputted locations; it cannot automatically detect where the lost item is.

6. **Limited Notifications: Unless advanced features like email or SMS alerts are added, users need to manually check the system for updates.**

13.CONCLUSION

The **Find My Stuff** project provides an efficient, reliable, and user-friendly solution for managing lost and found items. By leveraging modern web technologies such as **Python, Flask, HTML, CSS, JavaScript, and SQLite**, the system bridges the gap between people who lose their belongings and those who find them.

Unlike traditional manual methods, this web-based application centralizes all lost and found information, reducing time, effort, and confusion. Users can easily report lost or found items, browse existing records, and claim items with the help of clear and organized dashboards. The inclusion of images and detailed item descriptions further improves accuracy and item identification.

The system is designed to be **secure, responsive, and accessible**, making it suitable for everyday use in public spaces like colleges, offices, or community centers. While there are some limitations such as internet dependency and the need for manual verification, the project demonstrates a significant improvement over conventional processes.

Overall, **Find My Stuff** successfully achieves its goal of creating a digital platform that is practical, transparent, and helpful, promoting community cooperation and ensuring that lost items can be recovered in a timely and organized manner.

14.FUTURE SCOPE

The **Find My Stuff** project can be enhanced further to increase its efficiency, usability, and reach. Some possible improvements and extensions for the future include:

1. **Mobile Application:** Developing a mobile app version for Android and iOS to make the system more accessible on-the-go. Users can report or search for items directly from their smartphones.
2. **Real-Time Location Tracking:** Integration of GPS technology to automatically track the location where items are lost or found, improving accuracy and reducing manual errors.
3. **Advanced Notifications:** Adding features like **email alerts, SMS notifications, or push notifications** when a lost item is found or claimed. This ensures users stay informed in real-time.
4. **Cloud-Based Database:** Migrating from SQLite to a cloud-based database like **MySQL, Firebase, or PostgreSQL** to handle large-scale data, support multiple users, and enable faster access.
5. **AI and Image Recognition:** Implementing AI-based image recognition to automatically match lost and found items based on uploaded pictures, reducing dependency on manual description matching.
6. **Admin and Reporting Tools:** Adding analytics dashboards for admins to monitor trends, generate reports of lost/found items, and detect suspicious entries or misuse.
7. **Multilingual Support:** Making the platform available in multiple languages so users from different regions can use it easily.
8. **Community Integration:** Linking the platform with local institutions, offices, or public spaces to expand coverage and improve recovery rates of lost items.

15.REFERENCE

1.Lost & Found Web Application – IJRASET

https://www.ijraset.com/research-paper/lost-and-found-web-application

2.Web-Based Lost and Found System – ResearchGate

https://www.researchgate.net/publication/376856933_Web-Based_Lost_and_Found_System

3.Lost & Found Management System – JETIR

https://www.jetir.org/papers/JETIRGG06022.pdf

4.Lost and Found Application – ULM Journal (PDF)

https://klik.ulm.ac.id/index.php/klik/article/download/566/pdf